

The Transmission of Foreign Direct Investment (FDI) in International Trade: Accelerating the Manufacturing Sector

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(Academic Writing Sample)

Executive Summary

This study investigates three key research questions: (1) Does foreign direct investment (FDI) in Indonesia contribute positively to domestic output growth? (2) Is the resulting output primarily directed toward international markets or predominantly consumed domestically? (3) Which sector—manufacturing or services—exerts a greater influence on exports through FDI inflows? The analysis utilizes FDI and export data from Indonesia's Balance of Payments (NPI-BI) and quarterly output data from Indonesia's GDP statistics (PDB-BPS) covering the period from Q1 2010 to Q3 2024. The empirical approach employs a vector error correction model (VECM) and an autoregressive distributed lag (ARDL) model, preceded by unit root and co-integration tests.

The findings indicate that: (1) In the long run, FDI does not have a significant impact on aggregate exports. However, sectoral analysis reveals that manufacturing FDI positively affects manufacturing exports, while services FDI negatively affects services exports. (2) In the short run, FDI also shows no significant impact on overall exports, but services FDI exhibits a negative influence on services exports.

Based on these findings, several policy recommendations can be considered: (1) The government should prioritize incentives for foreign investment in the manufacturing sector over the services sector, as manufacturing FDI demonstrates a long-term export-enhancing effect, supporting an export-led growth strategy. (2) It is crucial to ensure that incoming investments yield positive spillovers to domestic firms through technology and knowledge transfer, human capital development, managerial capacity-building, and competitiveness enhancement. (3) Liberalizing foreign investment in the services sector should be approached cautiously to prevent a widening services trade deficit. Policies should focus on promoting services activities that support the manufacturing sector's competitiveness (i.e., the servicification of manufacturing).

Keywords: Exports, Foreign Direct Investment, Output

Introduction

President Prabowo has consistently reiterated Indonesia's commitment to a pro-foreign investment policy during his international visits. One of the most significant affirmations of this stance was conveyed at the APEC Summit on November 14, 2024, in Peru, where he emphasized that Indonesia's development-oriented approach to foreign capital inflows would remain a key priority in his administration's Quick Wins and the 2025–2029 National Medium-Term Development Plan (RPJMN).

Foreign direct investment (FDI) is most beneficial when it generates an optimal multiplier effect for the host country. A crucial policy transmission mechanism in this regard is ensuring that incoming investments are export-oriented rather than solely targeting the domestic market. When foreign investment primarily caters to domestic consumption, it can negatively impact Indonesia's primary income balance, as profits are repatriated abroad without significantly contributing to national trade revenues. In contrast, FDI directed towards export-oriented industries can enhance trade surplus, economic growth, and long-term financial sustainability.

As illustrated in Figure 1, Indonesia's FDI inflows have fluctuated over time but increased by 56.96% between 2010 and 2023. During the same period, Indonesia's exports grew by 101.18%, while

national output expanded by 81.47%. However, the data suggests that fluctuations in FDI, exports, and output have not always been aligned. This discrepancy highlights the need to strategically prioritize FDI inflows into sectors with the highest potential to boost exports.

Given that Indonesia, like any economic entity, operates under resource constraints, an optimal allocation of FDI becomes imperative. Aligning with President Prabowo's directive to maximize foreign investment, it is essential to assess the most impactful sectors for FDI. This analysis will focus on the manufacturing and services sectors, given their strategic role in driving Indonesia's export competitiveness and long-term economic resilience.

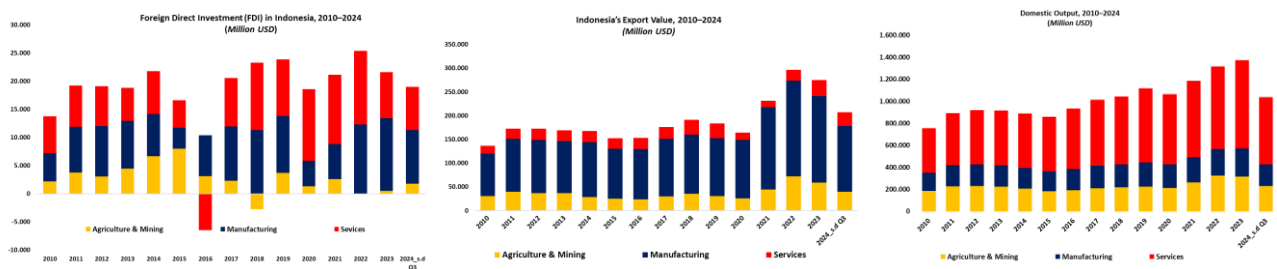


Figure 1: Indonesia's FDI, Exports, and Output, 2010–2024 (Million USD)

Source: Bank Indonesia, CEIC

Problem Statement

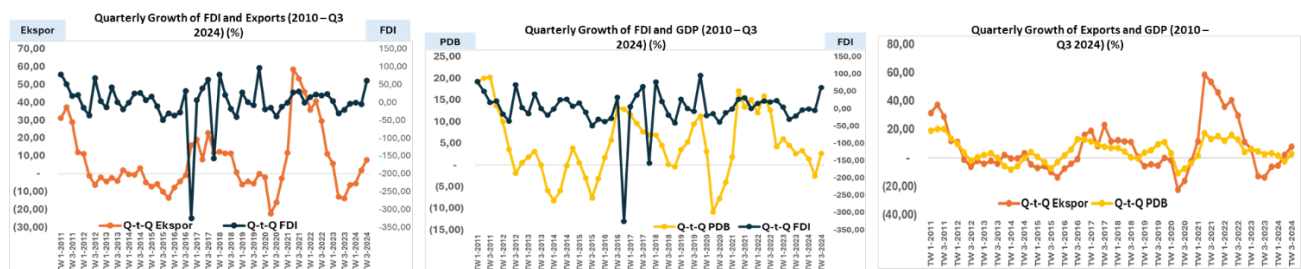


Figure 2: Fluctuations in FDI, Exports, and Output in Indonesia, 2010–2024 (Million USD)

Source: Bank Indonesia, CEIC.

Figure 2 illustrates the differing growth trends of FDI, output, and exports, characterized by high variability. The movements of FDI relative to exports and FDI relative to output exhibit distinct patterns, with quarterly FDI growth and export growth showing significant fluctuations. In certain periods, an increase in FDI is not necessarily accompanied by a corresponding rise in exports. Overall, the data indicate that the quarterly growth of FDI, output, and exports is highly volatile and does not always move in tandem. This pattern reflects the complexities of both global and domestic economic dynamics, as well as the various factors influencing foreign investment, production, and international trade. Descriptive statistical analysis suggests that the correlation between FDI, exports, and output in Indonesia is not consistently aligned. Therefore, further quantitative testing is required to conduct a more in-depth analysis of their interrelationships.

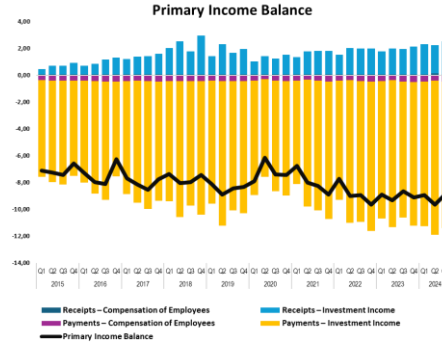


Figure 3: Primary Income Balance

Source: Bank Indonesia, CEIC.

Efforts to strengthen Indonesia's exports amid existing constraints require further examination to optimize key sectors. To achieve this, clear policy goals must be established to maximize foreign investment's role in promoting primary income balance (NPI) stability through export growth. Indonesia's trade balance remains volatile, with persistent deficits posing challenges. Additionally, limited financing and inadequate infrastructure for investment promotion further hinder export competitiveness. Non-export-oriented FDI has the potential to exacerbate the primary income deficit (Figure 3), as direct investment payments—especially equity capital returns—contribute significantly to the negative balance. Given this deficit, the economic benefits of FDI often extend more to foreign stakeholders than to Indonesia's economy. This raises key research questions:

1. Does FDI positively influence export growth in Indonesia?
2. Which sector—manufacturing or services—plays the most significant role in boosting exports?
3. What transmission mechanisms can strengthen FDI's contribution to exports?

Based on these research questions, several policy alternatives are formulated:

H1: If manufacturing FDI positively affects manufacturing exports, policies should prioritize attracting foreign investment in the manufacturing sector.

H2: If services FDI positively affects services exports, policies should focus on strengthening foreign investment in the services sector.

H3: If FDI in both sectors contributes to export growth, policies should aim to balance investment optimization across both sectors.

H4: If FDI in both sectors does not significantly impact exports, further exploration of other key sectors is necessary to identify more effective investment strategies.

To address these research questions, a quantitative analysis is conducted using econometric models, specifically the Vector Error Correction Model (VECM) and the Autoregressive Distributed Lag (ARDL) model. The empirical model specifications are as follows:

For the VECM approach:

$$\Delta EX_t = \alpha_3 + \sum_{i=0}^k \beta_{3i} \Delta PDB_{t-i} + \sum_{i=0}^k \gamma_{3i} \Delta FDI_{t-i} + \sum_{i=1}^k \theta_{3i} \Delta EX_{t-i} + \rho_3 ect_{t-1} + \varepsilon_{3t}$$

For the ARDL approach:

$$\Delta EX_t = \beta_{03} + \sum_{i=1}^p \alpha_{1i} \Delta EX_{t-1} + \sum_{i=0}^q \alpha_{2i} \Delta FDI_{t-1} + \sum_{i=1}^s \alpha_{3i} \Delta PDB_{t-1} + \delta_{13} EX_{t-1} + \delta_{23} FDI_{t-1} + \delta_{33} PDB_{t-1} + \varepsilon_{3t}$$

The choice of model depends on whether cointegration exists among the variables.

Recommendations

The Johansen cointegration test results indicate that models incorporating total FDI and manufacturing sector FDI were analyzed using the Vector Error Correction Model (VECM), while models including service sector FDI were tested using the Autoregressive Distributed Lag (ARDL) model (Appendix 2).

The long-term VECM analysis reveals that while total FDI does not significantly impact exports, FDI in the manufacturing sector positively affects manufacturing exports. According to Table 2, the P-value of 4.18 exceeds the absolute critical value of 2.0, confirming a statistically significant relationship. A 1% increase in manufacturing FDI leads to a 1.6% rise in manufacturing exports, demonstrating the sector's crucial role in boosting Indonesia's trade performance.

The short-term VECM analysis also confirms a positive impact of manufacturing FDI on manufacturing exports, with a notable lag effect. Specifically, current-period exports are significantly influenced by FDI from two previous periods. The P-value of 2.1, exceeding the absolute critical value of 2.0, supports this finding (Appendix 3).

Conversely, the ARDL model results indicate that both in the short and long run, FDI in the service sector negatively affects service exports. A rise in service sector FDI in the previous period correlates with a decline in current-period service exports. However, an increase in GDP from the service sector positively influences service exports in the same period and the previous period, suggesting that domestic economic expansion, rather than foreign investment, plays a more significant role in driving service exports. The model's coefficient of determination (R^2) is 36.3%, indicating moderate explanatory power (Appendix 4).

Table 1: Long-Term VECM Test Results

Time Series A: All Sectors

Variable	Coefficient	Standard Error	P-Value
Constant (C)	388.2429	-	-
Total Exports	1.00000	-	-
Total FDI	-3.441052	0.46625	-7.38031
Total GDP	-0.315719	0.08105	-3.89559

Time Series B: Manufacturing Sector

Variable	Coefficient	Standard Error	P-Value
Manufacturing Exports	1.00000	-	-
Manufacturing FDI	1.666752	-398.103	4.18674*
Manufacturing GDP	-4.462639	-113.766	-3.92264

Source: Data processed using EViews

The results demonstrate a consistent relationship between the manufacturing and service sectors. Thus, it can be concluded that both in the short and long run, FDI in the manufacturing sector positively influences manufacturing exports. Based on these findings, the following policy recommendation is proposed:

H1: Prioritizing efforts to attract foreign direct investment (FDI) in the manufacturing sector.

To implement this policy, several concrete measures should be undertaken:

1. Enhancing Government Support through Fiscal and Non-Fiscal Incentives for Foreign Investors in the Manufacturing Sector.

Fiscal and non-fiscal incentives, such as tax breaks and streamlined licensing procedures, can significantly enhance the attractiveness of the manufacturing sector to foreign investors. Foreign direct investment (FDI) in this sector facilitates the establishment of internationally oriented production facilities, expands export-oriented manufacturing capacity, and grants access to advanced technology and specialized expertise. In the long run, these measures contribute to strengthening Indonesia's export competitiveness, particularly for high-value-added manufactured goods.

The Ministry of Industry (Kemenperin) plays a pivotal role in formulating industrial policies, including those related to the manufacturing sector. The Ministry of Finance (Kemenkeu) is responsible for fiscal policies and tax incentives, while the Investment Coordinating Board (BKPM) is tasked with facilitating and coordinating foreign investment inflows. Effective coordination among these institutions is essential to ensure a cohesive and investor-friendly policy framework.

Regulations such as Law No. 25 of 2007 on Investment serve as the legal foundation for providing incentives to foreign investors. However, consistent implementation and clear regulatory guidelines are critical to preventing investment bottlenecks and fostering a stable and predictable business environment.

2. FDI inflows should prioritize export-oriented investments to drive higher economic growth through an export-led growth strategy.

Emphasizing export-oriented investment can strengthen the manufacturing sector by stimulating the production of goods destined for international markets. This approach supports export market diversification, reducing Indonesia's reliance on raw commodity exports. Additionally, fostering an export-driven manufacturing sector enhances the adoption of advanced skills and technologies in production, leading to improved product quality and global competitiveness.

The Ministry of Industry (Kemenperin) plays a crucial role in guiding the manufacturing sector toward export orientation while facilitating improvements in production quality and efficiency. The Investment Coordinating Board (BKPM) is responsible for ensuring that incoming FDI aligns with export-oriented objectives. Furthermore, the Ministry of Trade plays a significant role in expanding market access for Indonesian export products.

Policies and programs supporting export-driven industries, such as incentives for manufacturing exports and regulations under the Trade and Industry Law, can accelerate the attraction of export-oriented investments, reinforcing Indonesia's position in the global value chain.

3. The government must ensure that incoming investments generate positive spillover effects for local firms through technology transfer, managerial knowledge sharing, and workforce development, ultimately enhancing industrial competitiveness.

Effective technology transfer enables local firms to adopt more efficient production methods and utilize higher-quality raw materials, directly improving the quality and competitiveness of manufactured exports. Additionally, the transfer of managerial expertise enhances business efficiency, positioning local companies more effectively in global markets.

The Ministry of Industry (Kemenperin) plays a vital role in facilitating technology transfer programs and workforce upskilling initiatives. Collaboration with the Ministry of Manpower and Transmigration is essential to ensure that skill development programs align with industry needs. Furthermore, research institutions such as the Indonesian Institute of Sciences (LIPI) contribute to the development of industry-relevant technologies through research and innovation.

Legal frameworks such as Law No. 3 of 2014 on Industry and Law No. 13 of 2003 on Manpower support technology transfer and labor skill enhancement. However, a more integrated policy approach linking industrial development, vocational education, and access to new technologies is necessary to accelerate the effectiveness of technology transfer and strengthen Indonesia's industrial competitiveness.

4. Managing Foreign Investment in the Services Sector to Prevent a Widening Services Account Deficit.

The liberalization of foreign investment in the service sector must be approached cautiously to prevent an excessive service balance deficit. Investment inflows should prioritize the manufacturing sector over services, as manufacturing-driven FDI has stronger long-term export potential. If foreign direct investment (FDI) focuses solely on domestic market expansion, it may negatively impact the primary income balance, as profits generated by foreign-owned service firms tend to flow abroad.

A well-developed service sector can strengthen other areas of the economy, such as trade and logistics, which are crucial for supporting goods exports. However, excessive FDI in services without proper management may worsen the service balance deficit, given that earnings from this sector often accrue to foreign entities. Therefore, policy efforts should prioritize service industries that enhance domestic economic growth and export capacity, particularly financial services, logistics, and information technology.

The Coordinating Ministry for Economic Affairs (Kemenko Perekonomian) plays a central role in ensuring that foreign investment in the service sector remains balanced and does not undermine Indonesia's macroeconomic stability. The Ministry of Trade and the Investment Coordinating Board (BKPM) are also responsible for regulating service sector investments to prevent potential negative consequences for the domestic economy. Regulatory frameworks governing foreign investment in services are outlined in Law No. 25 of 2007 on Investment, along with sector-specific regulations that define entry conditions and restrictions for foreign investors in various service industries.

The effective implementation of FDI policies can significantly enhance Indonesia's export competitiveness. The government must ensure that incoming investments not only boost production capacity but also drive innovation, efficiency, and national economic stability. A coordinated approach among institutions is essential to align FDI policies with broader economic objectives, reinforcing Indonesia's position in global supply chains. The strategies outlined above are interconnected and mutually reinforcing. A comprehensive policy framework that integrates these approaches will maximize the benefits of foreign investment while ensuring sustainable economic growth. The transformation of these strategic policy recommendations is illustrated in the following figure.

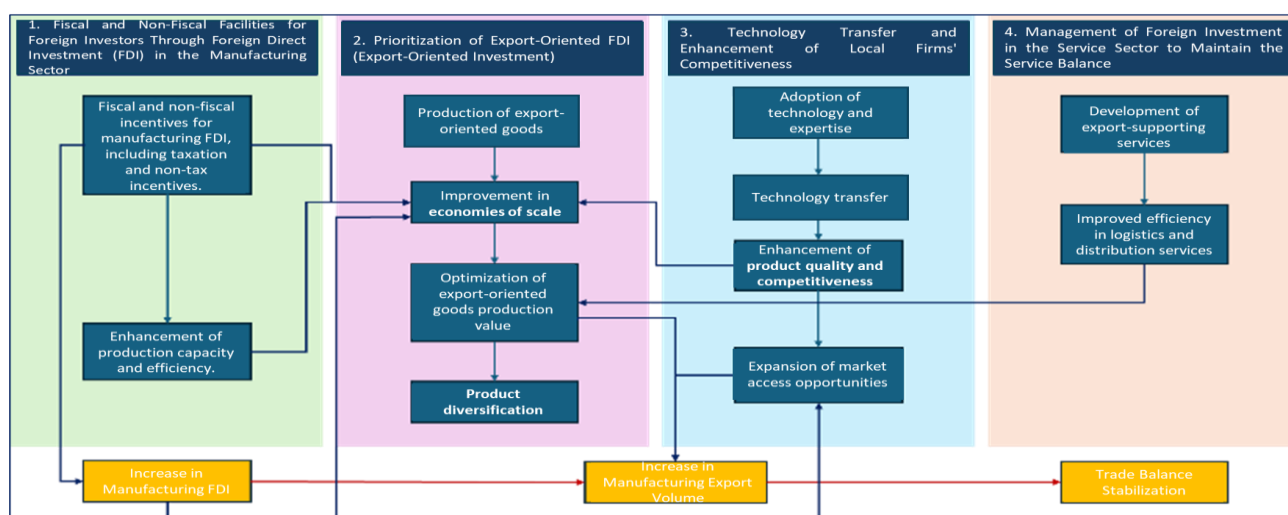


Figure 4: Policy Strategy Transformation Framework

Source: Internal Analysis

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Appendix

Appendix 1: Stationarity Test Results

- Stationarity Test for Aggregate Variables
 - The total FDI variable is stationary at the level.
 - The total export and total GDP variables are stationary at the first difference.
 - Since the stationarity levels differ—some variables are stationary at the level, while others at the first difference—the appropriate model for analyzing aggregate variables may involve either the Autoregressive Distributed Lag (ARDL) or the Vector Error Correction Model (VECM). A cointegration test is required to determine the suitable approach.
- Stationarity Test for the Manufacturing Sector Variables
All variables used in the analysis are stationary at the first difference. A cointegration test is necessary to determine whether the ARDL or VECM model should be applied.
- Stationarity Test for the Services Sector Variables
 - The service export and service GDP variables are stationary at the first difference.
 - The service FDI variable is stationary at the level.
 - Given the mixed stationarity levels—some variables are stationary at the level, while others at the first difference—the appropriate model for analyzing service sector variables may involve either the ARDL or VECM. A cointegration test is needed to confirm the appropriate model selection.

Variabel-	P value Level	P value 1 st Diff.	P value 2 nd Diff.
Time Series A : Keseluruhan Sektor			
Ekspor total	0.8383	0.0000*	-
FDI total	0.0000*	0.0000*	-
PDB total	0.9635	0.0002*	-
Time Series B : Sektor Manufaktur			
Ekspor Manufaktur	0.8167	0.0000*	-
FDI Manufaktur	0.2028	0.0000*	-
PDB Manufaktur	0.8916	0.0000*	-
Time Series C : Sektor Jasa			
Ekspor Jasa	0.1860	0.0000*	-
FDI Jasa	0.0000*	0.0000*	-
PDB Jasa	0.8322	0.0000*	-

Appendix 2: Cointegration Test Results

- Cointegration Test for Aggregate Variables
The test results indicate the presence of three cointegrating equations at a 0.05 significance level. Based on this analysis, the equation is modeled using the Vector Error Correction Model (VECM) with data at the first difference level. The lag length test suggests that the optimal model is at lag 2.
- Cointegration Test for Manufacturing Sector Variables
The test results indicate one cointegrating equation at a 0.05 significance level. Consequently, the equation is modeled using VECM with data at the first difference level. The lag length test suggests that the optimal model is at lag 4.
- Cointegration Test for Services Sector Variables
The test results indicate no cointegrating equations at a 0.05 significance level. Based on this analysis, the Autoregressive Distributed Lag (ARDL) model is deemed appropriate for assessing the effects of service sector FDI and service sector GDP on service exports.

Model	Prob. Crit. value
Time Series A : Keseluruhan Sektor	
Ekspor, FDI, dan PDB Total	None 0.0000*
	At most 1 0.0000*
	At most 2 0.0000*
Time Series B : Sektor Manufaktur	
Ekspor, FDI, dan PDB Manufaktur	None 0.0240*
	At most 1 0.2136
	At most 2 0.6376
Series C : Sektor Jasa	
Ekspor, FDI, dan PDB Jasa	None 0.2069
	At most 1 0.8200
	At most 2 0.2213

Appendix 3: Short-Term VECM Test Results

	d(Ekspor)	d(FDI)	d(PDB)	Interpretasi
Time series A: Seluruh Sektor, lag= 2				
d(Ekspor-1)	[-3.13244]	[-4.78728]	[-2.35340]	Pada Uji VECM jangka pendek Seluruh Sektor (Time Series A) : FDI periode t dipengaruhi oleh FDI 2 periode sebelumnya, PDB satu periode sebelumnya, dan PDB 2 periode sebelumnya.
d(Ekspor-2)	[-1.55008]	[-2.59429]	[-1.80493]	
d(FDI-1)	[-1.41473]	[2.95744]	[2.02103]	
d(FDI-2)	[-0.03306]	[2.13716]*	[1.71228]	
d(PDB-1)	[0.70226]	[4.54500]*	[-0.83810]	
d(PDB-2)	[0.38437]	[2.88881]*	[-1.80573]	
Time series B: Sektor Manufaktur, lag= 4				
d(Ekspor-1)	[-4.03876]	[0.25511]	[-1.16065]	Pada Uji VECM jangka pendek Sektor Manufaktur (Time Series B) : Ekspor periode t dipengaruhi oleh FDI pada 2 periode sebelumnya.
d(Ekspor-2)	[-2.40176]	[0.25006]	[-0.38020]	
d(Ekspor-3)	[-1.38774]	[1.07888]	[-0.38289]	
d(Ekspor-4)	[-0.15485]	[0.28674]	[0.15423]	
d(FDI-1)	[1.89122]	[0.19534]	[-1.75411]	
d(FDI-2)	[2.14886]*	[0.20200]	[-1.22618]	
d(FDI-3)	[1.84245]	[-0.32739]	[-0.86304]	
d(FDI-4)	[1.89129]	[0.33429]	[-0.50596]	
d(PDB-1)	[-1.27170]	[-2.01368]	[0.10059]	
d(PDB-2)	[-0.88233]	[-0.41315]	[-0.32227]	
d(PDB-3)	[-2.34048]	[-3.78113]	[-1.78377]	
d(PDB-4)	[0.37705]	[-1.91850]	[-0.28488]	

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Titik, Author, Publications Link	Tujuan	Metode	Hasil
- Penelitian Model Asing Ljungberg, Ekspor dan Pertumbuhan Ekonomi di Indonesia Analisis ARDL - ECM. - Cheng Chen, Lin, Aiguo Fan, and Economics Research, Vol. 10, Issue 3 https://doi.org/10.4236/economics.2019.103003 https://www.iiste.org/journals/article.php?journal_id=104&article_id=104003	Tujuan dari studi ini adalah untuk menganalisis hubungan antara FDI, ekspor, dan pertumbuhan ekonomi.	Penelitian ini menggunakan pendekatan Auto Regression Distributed Lag (ARDL) dan model ECM-ARDL untuk menguji hubungan kausalitas antara pertumbuhan ekonomi (PPN), ekspor, dan pertumbuhan ekonomi di Indonesia selama periode 1981-2016.	Hasil penelitian menunjukkan hubungan kausalitas dua arah antara arus masuk FDI dan pertumbuhan ekonomi, hubungan kausalitas satu arah antara ekspor kepada pertumbuhan ekonomi, dan tidak adanya hubungan kausalitas antara arus masuk FDI dan ekspor. Temuan penelitian ini memberikan bukti adanya pertumbuhan ekonomi yang didorong oleh FDI dan yang didorong oleh ekspor di Indonesia.

Appendix 8: Research Methodology.

1. Data Sources

- FDI data is obtained from the direct investment values recorded in the Capital and Financial Account of Indonesia's Balance of Payments, published by Bank Indonesia.
- Export data is derived from the trade values of goods and services in the Current Account of Indonesia's Balance of Payments.
- Output data represents the nominal domestic production value from the GDP by Production Sector, published by BPS (Statistics Indonesia).
- All data are measured in million USD.

2. Data Period

The study covers quarterly data from Q1 2010 to Q3 2024, based on data availability from CEIC.

3. Data Characteristics

The empirical analysis employs nominal quarterly data rather than natural logarithms, as some variables contain negative values.

4. Sectoral Classification

In addition to aggregate data analysis, this study conducts sectoral testing with the following classifications:

- **Agriculture and Mining Sector:** Includes food crops, plantations, livestock, forestry, fisheries, and mining.
- **Manufacturing Sector:** Covers industries such as basic metals; fabricated metal products (excluding machinery and equipment); chemicals and pharmaceuticals; motor vehicles and other transportation equipment; paper and printing; machinery, electronics, medical instruments, electrical equipment, precision optics, and watches; textiles; non-metallic minerals; leather products and footwear; rubber and plastics; wood; and other manufacturing industries.
- **Services Sector:** Includes electricity, gas, and water; transportation, warehousing, and telecommunications; housing, industrial estates, and office buildings; trade and repair services; hotels and restaurants; construction; and other service industries.

5. Model Selection

Based on the literature review, this study employs commonly used models to analyze the relationship between FDI, output, and exports: **Vector Error Correction Model (VECM)** and **Autoregressive Distributed Lag (ARDL)**.

6. Econometric Testing

- **Unit Root and Cointegration Tests:** Johansen's cointegration test is applied to determine the long-term stability of the variables.
- **Model Selection:**
 - If at least one cointegration relationship is found, the **VECM** model is used.
 - If the variables are stationary at different levels (some at level and others at first difference) and no cointegration is detected, the **ARDL** model is employed.